

IS6640: INFORMATION SYSTEMS PLANNING AND STRATEGY

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Information Systems Planning and Strategy

Subject Code

IS - Information Systems

Course Number

6640

Academic Unit

Information Systems (IS)

College/School

College of Business (CB)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

IS5642 Electronic Business Strategies and Management

IS6641 eCommerce Business Strategies and Management

Exclusive Courses

Nil

Part II Course Details

Abstract

The course aims to examine the role of information systems in assisting achievement of corporate objectives, and to consider methods for strategic and tactical planning of information systems.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Participate in developing a strategic plan for information systems development and assist in establishing an information technology policy.	20			
2	Use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools.	30	x	x	
3	Analyze a company's business model and capabilities, and determine the role for information systems in that organization.	30	x	x	x
4	Explain how decisions are made for the applications portfolio.	10			
5	Identify and assess the sourcing strategies and governance issues in information systems planning for an organization.	10			

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

	LTAs	Brief Description	CILO No.	Hours/week (if applicable)
1	LTA1: Lecture	Students will learn the following concepts of information systems strategies and information systems planning: - The value of IT in organizational strategic planning - IS use to create competitive advantages - Strategic alignment between business and IS strategies - Techniques for IS strategy formulation - scenario planning, five forces model, value chain concept, supply chain concept, ERP concept, and CRM concept - Strategy implementation and measurement - balanced scorecard approach - E-business models and related strategies - IS sourcing strategy - Managing IT investment - portfolio approach - Change management	1, 2, 3, 4, 5	Seminar/Workshop : 39 hours
2	LTA2: Class activities - team work	Students will involve in different types of class activities throughout the semester such as: - Real world cases studies: students will apply and integrate what they learn through classroom discussions and reading into real-world case analyses. - Discussions and presentations	1, 2, 3, 4, 5	

3	LTA3: Practical exercises - individual work	Students are required to carry out a strategic analysis for an organization. The individual assignment will be submitted around the middle of the semester. This exercise can then serve as the starting point for the group project which is submitted by the end of the semester.	2, 3	
4	LTA4: Team project - team work	On top of the case analyses done in the classes, students have to submit a group report assessing the effectiveness of IT uses in increasing the competitive advantages of a particular organization. Students will work in a team of 5-8 members. The team has to pick one organization as the study target. They are required to submit a business proposal on how IS/IT may affect the industry in which the organization is engaged and what business opportunities the organization should take in order to gain or sustain competitive advantage.	1, 2, 3, 4	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	AT1: Participation Students should participate in the class activities, such as, small group discussions and presentations, self reflection, raise and answer questions, and the like. The participation is used to assess students' understanding of the topics and their abilities to apply their skills.	1, 2, 3, 4, 5	10	-	Yes
2	AT2: Individual Assignment Analysis Students will analyze a real world case and write an essay to address relevant strategic issues.	2, 3	20	-	Yes
3	AT3: Group work A group should be responsible for finishing a group project (30%). The project which includes a project report and presentation will be allocated to let students analyze a topic in-depth and share their findings with other course participants. 20% will be contributed by the written report, while 10% will be counted on the presentation.	1, 2, 3, 4	30	-	Yes

Continuous Assessment (%)

60

Examination (%)

40

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

20

Additional Information for ATs

AT4: Final Examination

A written examination is developed to assess students' competence level of the taught subjects.

Note: Students must pass BOTH coursework and examination in order to get an overall pass in this course.

Assessment Rubrics (AR)

Assessment Task

AT1: Participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to participate in developing a strategic plan for information systems development and assist in establishing an information technology policy

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to analyze a company' s business model and determine the role for information systems in that organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to explain how decisions are made in an organization about the applications portfolio

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to identify and assess the sourcing strategies and governance issues in information systems planning for an organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Individual Assignment Analysis (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Individual Assignment Analysis (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to analyze a company' s business model and determine the role for information systems in that organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group work (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group work (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to analyze a company's business model and determine the role for information systems in that organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to participate in developing a strategic plan for information systems development and assist in establishing an information technology policy

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

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Excellent

(A+, A, A-) High

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(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to explain how decisions are made about the applications portfolio

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to identify and assess the effect of sourcing strategies and governance issues in information systems planning for an organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to participate in developing a strategic plan for information systems development and assist in establishing an information technology policy

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

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(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to analyze a company' s business model and determine the role for information systems in that organization

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Assessment Task

AT1: Participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to explain how decisions are made in an organization about the applications portfolio

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AT1: Participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

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Assessment Task

AT2: Individual Assignment Analysis (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

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Assessment Task

AT2: Individual Assignment Analysis (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

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Failure

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Assessment Task

AT3: Group work (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to use methods for strategic analysis, decide on the information needs and priorities of management in an organization, and assess information technology systems as competitive tools

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group work (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to analyze a company' s business model and determine the role for information systems in that organization

Excellent

(A+, A, A-) High

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(B-, C+, C) Basic

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Assessment Task

AT4: Final Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to participate in developing a strategic plan for information systems development and assist in establishing an information technology policy

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Criterion

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(B-, C+, C) Basic

Failure

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Part III Other Information

Keyword Syllabus

- Business strategy concept
- IS strategy concept
- Strategic alignment between business and IS strategies
- Emerging information technology opportunities including Digital Transformation and Artificial Intelligence
- Scenario planning
- Five forces model, value chain concept, supply chain concept, enterprise resource planning concept, customer relationship management concept
- eBusiness models and strategies
- Balanced scorecard analysis
- Application portfolio approach
- Sourcing strategy
- IS outsourcing

Reading List**Compulsory Readings**

Title	
1	Nil

Additional Readings

Title	
1	Applegate, L., Austin, R.D. and Soule, D.L., Corporate Information Strategy Management: Text and Cases, 8th edition, McGraw Hill, 2009.
2	Applegate, L., Austin, R.D. and McFarlan, F., Corporate Information Systems Management: Text and Cases, 7th edition, McGraw Hill, 2007.
3	Pearlson K., Saunders, C., Managing and Using Information Systems - A Strategic Approach, 3rd edition, John Wiley & Son, 2006
4	Luftman J., et al., Managing the Information Technology Resource - Leadership in the Information Age, Pearson Prentice Hall, 2004.
5	Supplementary readings may be assigned for each lecture